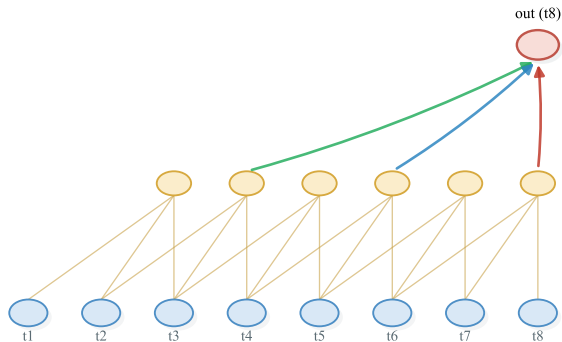


Inductive ST-GNN building blocks

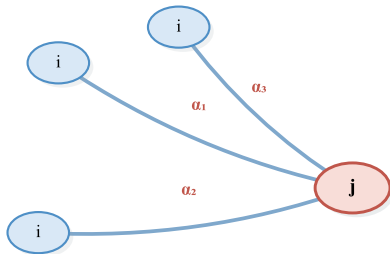
(a) TemporalConv — dilated causal conv (per node)

kernel 3, dilation 1 then 2; left-padded + causal trim (no future leak)



parameters: $(\text{in} \cdot \text{out} \cdot 3)$ — independent of $|V|$

(b) GAT layer — edge-weighted attention



$$e_{ij} = \text{LeakyReLU}(a_{\text{src}} \cdot Wx_i + a_{\text{dst}} \cdot Wx_j) + \log w_{ij}$$

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) \quad h_j = \sum_i \alpha_{ij} \cdot Wx_i$$

parameters: $\text{out} \cdot (\text{in}+2)$ — no per-node term, independent of $|V|$